

gRPC MACH Architecture

TCO Networking Costs Calculator

Point-to-Point Integration Pattern · Protocol Buffer IDL · AWS Egress Pricing

✔ File 'complexUnaryRPCPattern.proto' uploaded successfully!



Phase 4 — TCO Networking Costs Report

Full cost breakdown including architecture tactics, unit economics, and stakeholder summary.

SERVICE UNDER ANALYSIS

order-management-system

O M S

🔌 BUC1

⚡ 1200 base RPS

👥 450 End Users

📁 —

APPLIED ARCHITECTURE TACTICS

None selected — base configuration

MESSAGE TYPE	PROTO SIZE (BYTES)	EFFECTIVE SIZE (BYTES)	REQUESTS / MONTH	THROUGHPUT (GB/MONTH)	TRANSFER COST (USD/MONTH)
Request	8025	8025	3B	23,246.7055 GB	—
Response	794	794	3B	2,300.0479 GB	\$ 207.00

🗃 Total Cost of Ownership (TCO) Breakdown

Cost Component	DR / BC	Monthly Cost	% of TCO
AWS Egress (Response Transfer)	Networking	\$207.00	73.9%

Cost Component	DR / BC	Monthly Cost	% of TCO
Cloud Infrastructure	Cloud + Container	\$73.00	26.1%
Containerized Cluster \$73.00			
Total Cost of Ownership (TCO)		\$280.00/mo	100.0%

* % of TCO is calculated against the sum of ALL cost rows shown above (must total 100%). DR = Disaster Recovery | BC = Business Continuity.

 Unit Economics

TOTAL MONTHLY TCO (NETWORKING + CLOUD INFRA)

\$280.00

\$207.00 egress + \$73.00 cloud infra

 Per-Microservice Cost Metrics

TCO / MICROSERVICE / MONTH

\$280.00

This service - full TCO

COST / REQUEST

\$9.0021e-8

3.110.400.000 req/mo

ANNUAL PROJECTION

\$3360.00

12x monthly TCO

 Per-End-User Cost Metrics (450 end users)

TCO / END USER / MONTH

\$0.6222

\$280.00 ÷ 450 users

TCO / END USER / DAY

\$0.020741

Monthly ÷ 30

BUSINESS USE CASE

—

Pattern cost attribution

 Revenue, ARPU & ROI Analysis

Revenue model: The value entered in Phase 1 is the *expected monthly revenue per end user* for this service. Total monthly revenue = that value × number of end users. **ARPU** = revenue per user per month. **ROI** = (total revenue – TCO) ÷ TCO. **Break-even users** = minimum end users needed to cover full TCO at current ARPU. Net Margin per user = ARPU – TCO per user. Present to Finance and Leadership to frame architecture cost as a revenue-generating investment.

TOTAL MONTHLY REVENUE

\$4050

ARPU / MONTH

\$9

NET MARGIN / USER / MO

\$8.3778

$\$9.00/\text{user} \times 450 \text{ users}$ Revenue per end user /
month

ARPU - TCO/user

MONTHLY ROI**1346.4%**

(Revenue - TCO) ÷ TCO

ANNUAL ROI**1346.4%**

12× monthly projection

BREAK-EVEN USERS**32**

users to cover full TCO

REVENUE / \$1 INFRA**\$14.46**

Revenue efficiency

ANNUAL REVENUE**\$48.600**

12× monthly revenue

FinOps Architecture Notes & Disclaimers

1 — IP Address Planning (CIDR Sizing)

The more redundancy you build into AWS (Multi-AZ, replicas, NAT Gateways, ENIs), the more private IP addresses are consumed. Plan your VPC CIDR block generously (e.g. /16 = 65,536 IPs) and **leave enough address space within each subnet CIDR for auto-scaling headroom**. A /24 subnet provides only 251 usable IPs — exhausted quickly in a multi-tier, auto-scaled architecture.

2 — Multi-Tier Architecture in AWS Terms

Separating MACH components (frontend, API gateway, gRPC services, databases) into **different subnets** makes this a **multi-tier architecture** in AWS terms. Each tier needs dedicated subnets with its own Security Groups and NACLs.

3 — Intra-VPC Connectivity

Use **VPC Endpoints** (PrivateLink) to access AWS services without internet routing; **Transit Gateway** for cross-VPC connectivity; **VPC Flow Logs** for visibility. Each component in a separate subnet requires explicit route table entries.

* Outbound (response) data transfer cost uses AWS standard internet egress tiered pricing for US East (N. Virginia): first 10 TB at \$0.09/GB, next 40 TB at \$0.085/GB, next 100 TB at \$0.07/GB, beyond 150 TB at \$0.05/GB. Inbound (request) traffic is free. Effective sizes include TLS (RFC 8446) and JWT (RFC 7519) overhead where applicable. [AWS Pricing Calculator](#) ↗



Phase 5 — Portfolio Summary

Aggregated TCO for all modelled microservices. Data persisted in browser local storage.

i Each service you calculate and add to the portfolio appears here with its description, Business Use Case, and total monthly TCO. Use **Add to Portfolio** in Phase 4. Data is stored in your browser’s **local storage** — no server required, survives page reload.

MACH Architecture — TCO Portfolio Summary

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Service	BUC	Purpose	Monthly TCO	Annual TCO
order-m-s	BUC4		\$100.00/mo	\$1200.00/yr
order-management-system	BUC1	O M S	\$926.68/mo	\$11120.16/yr
order-retriever-system	BUC4		\$2935.13/mo	\$35221.56/yr
TOTAL (3 services)			\$3961.81/mo	\$47541.72/yr

Architectural TCO estimate. Validate with Engineering, FinOps, Finance and Procurement before acting on these figures.